
Microeconomics I

Class 3 — Utility representation

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Plan for today

Five stars or a thumb

Utility functions: what the numbers mean

Continuity, existence, the lexicographic case

Break

Quasiconcavity and concavity

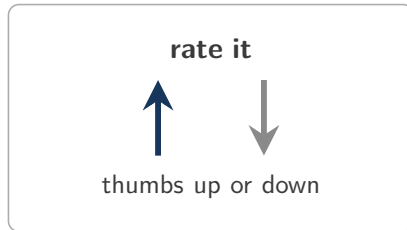
Common utility functions

Summary; problem set 1

Practicalities

- Problem set 1 today, on paper: one copy per group, your group's own numbers
- Due in class Wed 14 Oct, one paper copy per group, all names on it
- Office hours: Wednesdays 12:00–13:00, after class
up to 20 minutes per student; those who book by e-mail have priority
- Slides and reading: pkocourek.com/teaching/micro-1 · pkocourek@ub.edu

Five stars or a thumb?



Which of the two is a utility function?

Two rating screens of a real streaming service; what it did with them, and why, given in class.

Revealed in class

Before Tuesday: which screen tells the service more about what you like, and which one is easier to answer honestly? Are those the same question?

Definition: a utility function represents $\succsim$

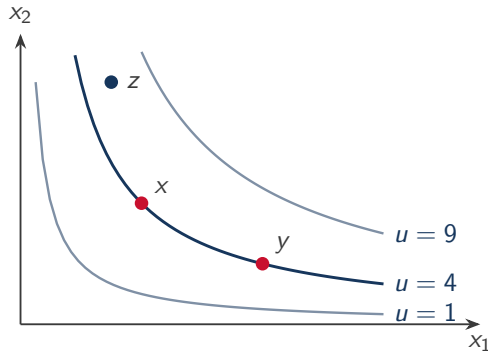
$u : X \rightarrow \mathbb{R}$ **represents** $\succsim$ if, for all $x, y \in X$,

$$x \succsim y \iff u(x) \geq u(y)$$

Represented $\Rightarrow \succsim$ is rational

complete and transitive, because $\geq$ on $\mathbb{R}$ is

MWG Definition 1.B.2, Proposition 1.B.2.



$u(x) = x_1 x_2$: $u(x) = u(y) = 4$, so $x \sim y$; $u(z) = 6$, so $z \succ x$.

From class 2: four of the five were already “ranked by a number”

	$x \succsim y \iff$	$u(x) =$
perfect substitutes	$x_1 + x_2 \geq y_1 + y_2$	$x_1 + x_2$
perfect complements	$\min\{x_1, x_2\} \geq \min\{y_1, y_2\}$	$\min\{x_1, x_2\}$
Cobb–Douglas type	$x_1 x_2 \geq y_1 y_2$	$x_1 x_2$
quasilinear	$x_1 + \sqrt{x_2} \geq y_1 + \sqrt{y_2}$	$x_1 + \sqrt{x_2}$
lexicographic	$x_1 > y_1$, or $x_1 = y_1$ and $x_2 \geq y_2$	?

Three questions for today: is the number **unique**? When does one **exist**? What do its **properties** say about $\succsim$?

In pairs: same preference as $u(x) = x_1x_2$?

$v(x) =$ same $\succsim$?

$$\sqrt{x_1x_2}$$

$$2 - \frac{1}{1 + x_1x_2}$$

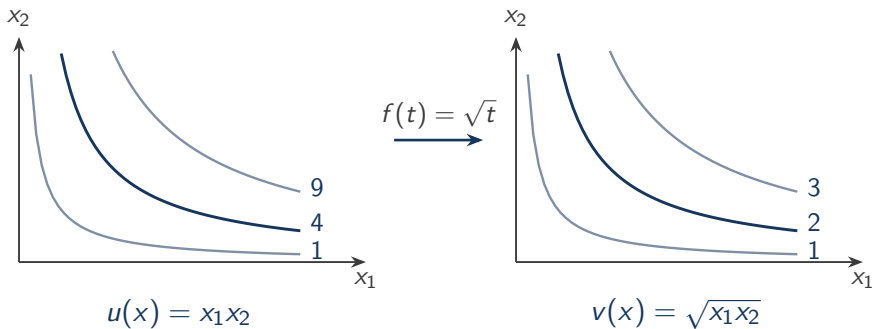
$$x_1 + x_2$$

$$x_1x_2^2$$

Four minutes. Same preference means: the same ranking of every pair of bundles.

Answers revealed in class. Try it in pairs before Tuesday: for a “no”, find two bundles ranked differently.

The numbers are labels: any strictly increasing relabelling will do



u represents $\succsim$ and $f : \mathbb{R} \rightarrow \mathbb{R}$ strictly increasing $\Rightarrow v = f \circ u$ represents $\succsim$

Utility is **ordinal**: same map, different numbers on the curves. Differences of u mean nothing.

MWG 3.C, after Proposition 3.C.1; Section 1.B. Proof: $v(x) \geq v(y) \iff u(x) \geq u(y) \iff x \succsim y$.

Back to the stars

stars invite arithmetic

★★★★★ + ★★★☆☆ = 3?

averages and gaps: a **cardinal** use

a thumb

liked it, or not

nothing to add up: **ordinal** only

The theory needs only the ranking. Nothing in $\succsim$ says how much better.

Revealed in class: what the service's members did with its stars.

Continuity and existence

MWG 3.C

Definition: continuous preferences

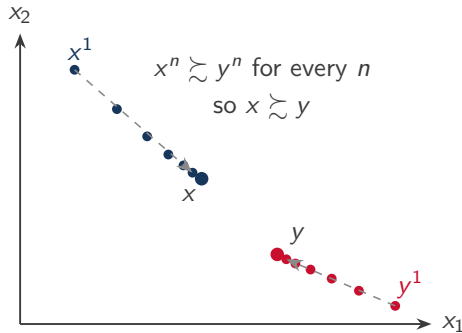
$\succsim$ on X is **continuous** if it is preserved under limits:

$$x^n \succsim y^n \text{ for all } n,$$

$$x = \lim x^n, \quad y = \lim y^n$$

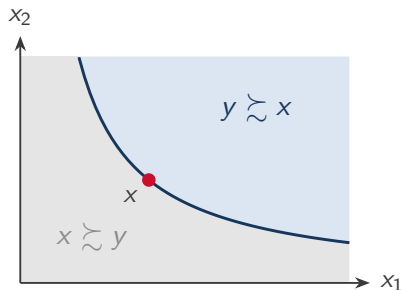
$$\Rightarrow x \succsim y$$

MWG Definition 3.C.1.



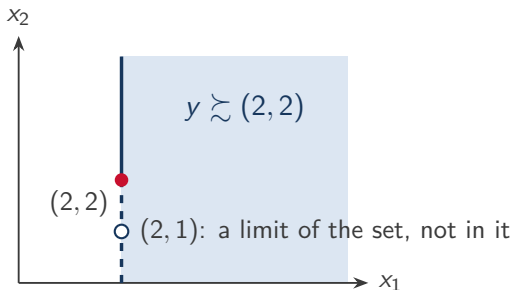
No sudden reversal at the limit: what holds along the whole sequence holds at the end.

Equivalently: upper and lower contour sets are closed



continuous

the curve belongs to both sets: closed



lexicographic

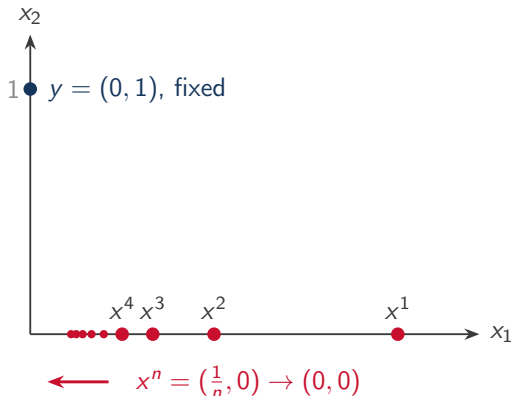
the dashed edge is missing: not closed

Closed: limits of points of the set stay in the set.

Right: $(2 + \frac{1}{n}, 1) \succsim (2, 2)$ for every n , but the limit $(2, 1)$ is not in the set.

MWG 3.C, text after Definition 3.C.1 (one direction; the converse is Exercise 3.C.3).

Lexicographic preferences reverse at the limit



$x^n \succ y$ for every n

$\frac{1}{n} > 0$: more of good 1 decides

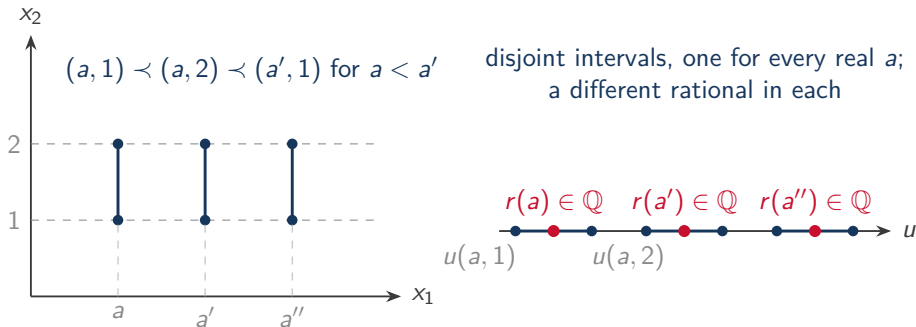
$\lim x^n = (0, 0) \prec (0, 1) = y$

tie in good 1: now good 2 decides

Not continuous.

The ranking flips exactly where the first coordinates become equal. Example 3.C.1, continued.

Lexicographic preferences have no utility function at all



A one-to-one map from the reals into the rationals: uncountable into countable. Impossible.

MWG Example 3.C.1, small print on p. 46. This rules out every u , continuous or not.

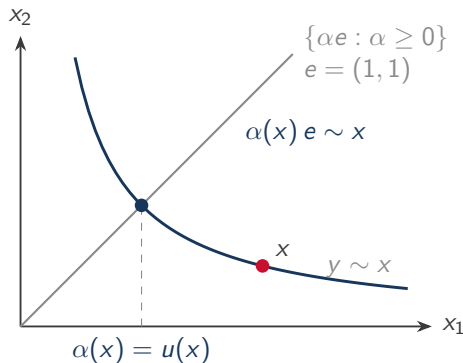
Continuity is enough: a continuous utility function exists

$\succsim$ rational and continuous on X
 $\Rightarrow$ some continuous $u : X \rightarrow \mathbb{R}$
represents $\succsim$

Proof idea, $\succsim$ monotone:

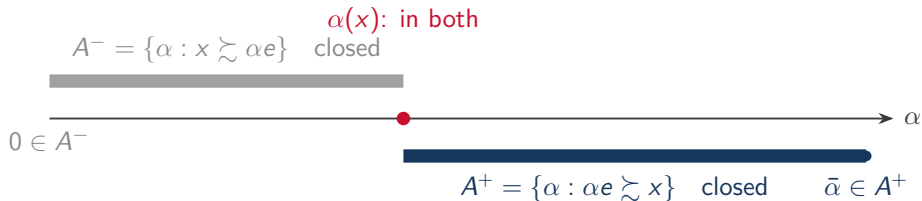
$$u(x) = \alpha(x) \text{ with } \alpha(x) e \sim x$$

MWG Proposition 3.C.1, Figure 3.C.1 (monotone case).



Each curve is labelled by where it crosses the diagonal; further out crosses further along.

Why every curve crosses the diagonal, once



closed (continuous), nonempty (monotone), cover $\mathbb{R}_+$ (complete) $\Rightarrow$ they overlap

monotone: $\alpha' > \alpha \Rightarrow \alpha' e \succ \alpha e$, so they overlap in **one** point

$$u(x) \geq u(y) \iff \alpha(x)e \succsim \alpha(y)e \iff x \succsim y$$

MWG, proof of Prop. 3.C.1, pp. 47–48. $\mathbb{R}_+$ is connected: two nonempty closed sets covering it meet.

In pairs: continuous? represented by some u ?

$$x \succsim y \iff$$

continuous some u

$$x_1 x_2 \geq y_1 y_2$$

$$x_1 > y_1, \text{ or } x_1 = y_1 \text{ and } x_2 \geq y_2$$

$$\lfloor x_1 \rfloor \geq \lfloor y_1 \rfloor \quad (\text{whole units of good 1 only})$$

$$\|x - (5, 5)\| \leq \|y - (5, 5)\|$$

Six minutes. For “not continuous”, give a sequence that reverses at the limit.

Answers revealed in class. Try it in pairs before Tuesday.

Break

Ten minutes

Quasiconcavity and concavity

MWG 3.C

Definition: quasiconcave utility

u is **quasiconcave** if every set $\{y \in X : u(y) \geq u(x)\}$ is convex, that is,

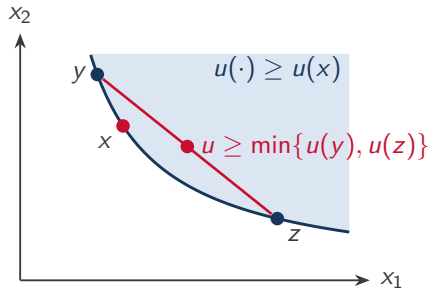
$$u(\alpha x + (1 - \alpha)y) \geq \min\{u(x), u(y)\}$$

for all $x, y \in X$ and $\alpha \in [0, 1]$; **strictly**: $>$ when $x \neq y$, $\alpha \in (0, 1)$

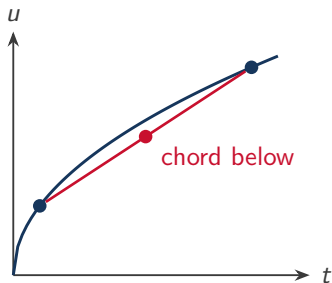
$\succsim$ convex $\iff u$ quasiconcave
strictly convex $\iff$ strictly quasiconcave

MWG 3.C, pp. 49–50.

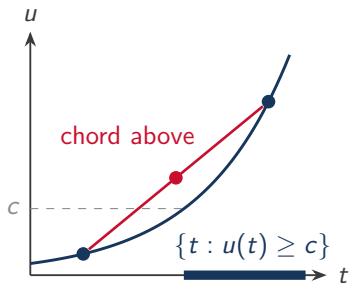
Class 2's picture: the upper contour sets of $\succsim$ are the sets $\{u \geq u(x)\}$.



Concave is stronger, and it is about the numbers



concave: $u(t) = \sqrt{t}$
and quasiconcave



quasiconcave, not concave: $u(t) = e^{0.6t}$
increasing, so every level set is a half-line: convex

concave: $u(\alpha x + (1 - \alpha)y) \geq \alpha u(x) + (1 - \alpha) u(y) \Rightarrow$ quasiconcave, not $\Leftarrow$
MWG 3.C, p. 50: some convex $\succsim$ have no concave representation at all.

What survives a strictly increasing relabelling $v = f \circ u$

	survives	
represents the same $\succsim$	✓	that is what f increasing means
increasing ($x \gg y \Rightarrow v(x) > v(y)$)	✓	monotone $\succsim$
quasiconcave	✓	same level sets, same $\succsim$: convex $\succsim$
the MRS at every bundle	✓	$f'(u) u_1 / f'(u) u_2 = u_1 / u_2$; f differentiable, $f' > 0$
concave	✗	$x_1 x_2$ vs $\sqrt{x_1 x_2}$: next slide
marginal utility $\partial u / \partial x_1$	✗	rescaled by f'
continuous	✗	a jump in f makes v jump

Top: **ordinal**, facts about $\succsim$. Bottom: **cardinal**, facts about one convenient u .

MWG 3.C, p. 50 and footnote 6.

In pairs: quasiconcave? concave?

$u(x) =$	quasiconcave	concave
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$x_1 x_2$		
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$\sqrt{x_1 x_2}$		
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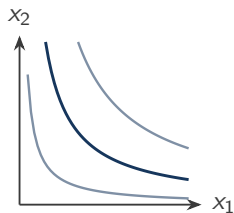
$x_1 + x_2$		
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$(x_1 + x_2)^2$		
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Five minutes. Test concavity at $(0, 0)$, $(2, 2)$ and their midpoint.

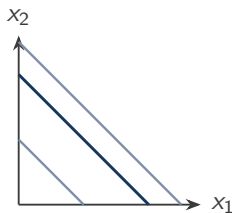
Answers revealed in class. Try it in pairs before Tuesday.

Four utility functions you will use all month



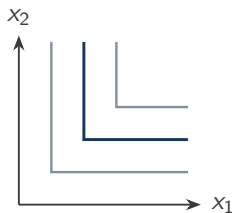
Cobb–Douglas

$$x_1^\alpha x_2^\beta$$
$$\text{MRS} = \frac{\alpha x_2}{\beta x_1}$$



perfect substitutes

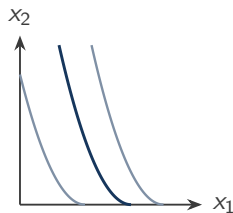
$$ax_1 + bx_2$$
$$\text{MRS} = \frac{a}{b}$$



Leontief

$$\min\{ax_1, bx_2\}$$

no MRS at the kink

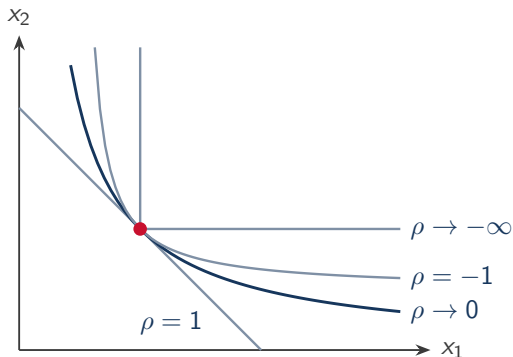


quasilinear

$$x_1 + \varphi(x_2)$$
$$\text{MRS} = \frac{1}{\varphi'(x_2)}$$

$\alpha, \beta, a, b > 0$; φ increasing and concave. $\text{MRS}_{12} = u_1/u_2$, class 2. Leontief kinks: MWG Figure 3.C.3.

One formula holds the whole family: CES



$$u(x) = \left(\frac{1}{2}x_1^\rho + \frac{1}{2}x_2^\rho \right)^{1/\rho}$$

$\rho \leq 1$, $\rho \neq 0$; for $\rho < 0$ take $x \gg 0$

$\rho = 1$: perfect substitutes $\frac{1}{2}(x_1 + x_2)$

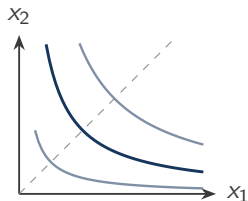
$\rho \rightarrow 0$: Cobb–Douglas $\sqrt{x_1 x_2}$

$\rho \rightarrow -\infty$: Leontief $\min\{x_1, x_2\}$

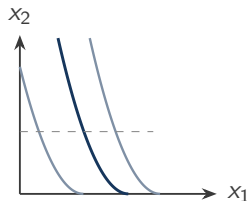
All four curves are $u = 2$. Lower ρ : a sharper bend, the goods substitute less easily.

Elasticity of substitution $\sigma = 1/(1 - \rho)$; Muñoz-García §1.8.4 (with weights a, b). Returns with demand.

Class 2's two families, in utility form



homothetic



quasilinear in good 1

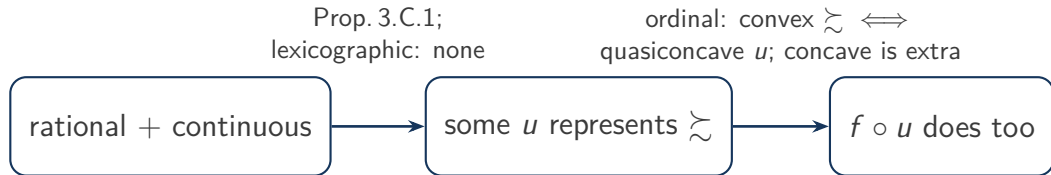
continuous $\succsim$: **homothetic** $\iff$ some u with $u(\alpha x) = \alpha u(x)$ for $\alpha > 0$
 $\sqrt{x_1 x_2}, \quad x_1 + x_2, \quad \min\{x_1, x_2\}$

continuous $\succsim$: **quasilinear** in good 1 $\iff$ some u of the form $x_1 + \varphi(x_2)$
 $x_1 + \sqrt{x_2}, \quad x_1 + \ln(1 + x_2)$

“Some u ”: a cardinal choice. $x_1 x_2$ is homothetic too, and not homogeneous of degree one.

MWG Exercise 3.C.5, stated on p. 50; quasilinear on $X = \mathbb{R} \times \mathbb{R}_+$. Both return with demand next week.

Today in one line



Wednesday: budget set, utility maximisation, Walrasian demand.

Problem set 1: one copy per group, due Wed 14 Oct.

Read MWG 2.D and 3.D (or Muñoz-García ch. 2) for Wednesday 7 October.